Draft version December 13, 2021
Typeset using LaTeX **twocolumn** style in AASTeX63

Korg: a modern 1D LTE spectral synthesis package

Adam Wheeler,[1] Matthew Abruzzo,[1] Andy Casey,[1] and Melissa Ness[1]

[1] *Department of Astronomy, Columbia University, Pupin Physics Laboratories, New York, NY 10027, USA*

ABSTRACT

We present Korg, a new package for 1D LTE (local thermal equilibrium) spectral synthesis. Korg can compute theoretical spectra from the near-ultraviolet to the near-infrared and implements both plane-parallel and spherical radiative transfer. Korg is compatible with automatic differentiation libraries, and easily extensible, making it ideal for statistical inference and parameter estimation applied to large data sets. We outline the inputs and internals of Korg, and compare its output spectra to those produced by other codes. We show that Korg is 1–30 times faster than other codes under typical use. Documentation and installation instructions are available at https://ajwheeler.github.io/Korg.jl/stable/.

Keywords: spectroscopy

1. INTRODUCTION

Improvements in instrumentation have yielded exponential growth in the amount of spectral data to analyse. Creating pipelines that can keep up with these data is nontrivial. There are several extant codes for 1D LTE spectral synthesis, including TurboSpectrum (Plez et al. 1993; Plez 2012), MOOG (Sneden et al. 2012), SYNTHE (Kurucz 1993; Sbordone et al. 2004), SME, (Valenti & Piskunov 1996, 2012), and SPECTRUM (Gray & Corbally 1994). While they have enabled a huge volume of research, these codes can be difficult to use for the uninitiated, and require input and output through custom file formats, impeding integration into analysis code. Here we present Korg, a new 1D LTE synthesis package, written in Julia and suitable for easy integration with scripts and use in an interactive environment. As the first such code in more than two decades, Korg benefits from modern numerical libraries not available at the time earlier packages were authored.

The assumption of LTE is that baryonic matter is described by thermal distributions, and radiation is only very slightly out of equilibrium. The source function is dominated by collisions of baryonic matter and is Planckian. While non-LTE (NLTE) calculations are important for producing the most unbiased possible model spectra, they are prohibitively slow, and not yet suitable for application that require computing many spectra. Fortunately, corrections to LTE level populations (e.g. Amarsi et al. 2020), equivalent widths, or abundances (e.g. Lind et al. 2011; Amarsi et al. 2015; Bergemann et al. 2012; Osorio & Barklem 2016) can be calculated from NLTE simulations. These can be applied to LTE codes to produce approximate NLTE spectra at relatively little computational cost. Additionally, biases from the LTE will roughly cancel for similar stars, and differential abundance estimates can have high accuracy.

The performance of spectral synthesis codes is most important when fitting observational data. Because synthesis must be embedded in an inference loop, the analysis of of a single spectrum may trigger tens or hundreds of syntheses. Even when many parameters may be estimated by interpolating over a precomputed grid of spectra (e.g. Holtzman et al. 2018; Buder et al. 2021), individual abundances are best done with targeted syntheses of individual lines, and generating the grid from which to interpolate can be computationally expensive. Inference and optimization can also be sped up by fast and accurate derivatives of the objective function, most easily produced via automatic differentiation. Korg is designed to be compatible with the automatic differentiation packages (e.g. ForwardDiff), which can provide derivative spectra in roughly the same amount of time required for a single synthesis (as discussed in section 4; see Figure 6).

Corresponding author: Adam Wheeler
a.wheeler@columbia.edu

2. DESCRIPTION OF CODE

KORG takes as inputs, a model atmosphere, a linelist, and abundances for each element in $A(X)$ form,[1] assumed to be constant throughout the atmosphere. By default, abundances follow the solar ratios from Asplund et al. (2009), and the abundances of metals can be set individually or uniformly scaled with a `metallicity` parameter.

2.1. *radiative transfer*

KORG solves the radiative transfer equation in a plane-parallel atmosphere,

$$\frac{\mu}{\alpha_\lambda}\frac{\mathrm{d}I_\lambda}{\mathrm{d}z} = S_\lambda - I_\lambda \quad , \tag{1}$$

or in a spherical atmosphere,

$$\frac{\mu}{\alpha_\lambda}\frac{\partial I_\lambda}{\partial r} + \frac{1-\mu^2}{\alpha_\lambda r}\frac{\partial I_\lambda}{\partial \mu} = S_\lambda - I_\lambda \quad , \tag{2}$$

where I_λ is the intensity of radiation at wavelength λ, S_λ the source function, α_λ the absorption coefficient, z the negative depth into the atmosphere, r, the distance to the center of the star, and μ the cosine of the angle between the r/z axis and the line of sight, to compute the disk-averaged intensity, i.e. the astrophysical flux,

$$\mathcal{F}_\lambda = 2\pi \int_0^1 \mu I_\lambda^{\mathrm{top}}(\mu)\,\mathrm{d}\mu \quad . \tag{3}$$

Here, $I_\lambda^{\mathrm{top}}(\mu)$ stands for $I_\lambda(z=0,\mu)$ in the plane-parallel case and $I_\lambda(r=R,\mu)$ in the spherical case, where R is the radius of the outermost atmospheric layer. The total flux from the star is then $F_\lambda = (R/d)^2\mathcal{F}_\lambda$, where d is the distance to the star.

In the plane-parallel case,

$$\mathcal{F}_\lambda = 2\pi \int_{z'}^0 S_\lambda(z) E_2(z)\alpha_\lambda \mathrm{d}z \quad , \tag{4}$$

where E_2 is the second order exponential integral and z' is the depth of the bottom of the atmosphere, but the spherical case admits no further analytic simplifications. When using a spherical model atmosphere, to obtain the astrophysical flux, $\mathcal{F}_\lambda$, KORG calculates the intensity, I_λ^{top} at a discrete grid of μ values, by integrating along rays from the lowest atmospheric layer to the upper layer for many of surface μ values, which is valid since the source function is isotropic under LTE. Rays which do not intersect the lowest atmospheric layer are cast from the far side of the star. Appendix A contains a few numerical details on the calculation of transfer integrals.

[1] $A(X) = \log_{10}(n_X/n_{\mathrm{H}})+12$ where n_X is the total number density of element X and n_{H} that of hydrogen.

Since KORG assumes that the stellar atmosphere is in LTE, S_λ is the blackbody spectrum. Most of the computation is then spent calculating the absorption coefficient, α_λ, which has contributions from both lines and continuum sources.

2.2. *chemical equilibrium*

In order to compute the absorption coefficient, α_λ, at each layer of the atmosphere, KORG must solve for the number density of each species (e.g. H I, Ca II, neutral CO). Treating the number density of each neutral atomic species as the free parameters, KORG uses the NLSOLVE library (Mogensen et al. 2020) to solve the system of Saha and molecular equilibrium equations with the temperature, total number density, and number density of free electrons set by the model atmosphere, and the total abundance of each element set by the user. By default, KORG uses atomic ionization energies, atomic partition functions, molecular equilibrium constants from Barklem & Collet (2016) (the ionization energies are originally from Haynes 2010), but alternatives can be passed by the user.

By default, KORG solves the equilibrium equations taking into account all elements up to uranium, as well as 247 diatomic molecules. As they exist in extremely small quantities in LTE atmospheres, KORG neglects species which are more than doubly ionized. It also neglects ionic and triatomic molecules at present, although we plan to address this in a future version.

2.3. *Line absorption*

The contribution of each line to the total absorption coefficient, α_{line}, is

$$\alpha_\lambda^{\mathrm{line}} = \sigma n \frac{\exp(-\beta E_{\mathrm{up}}) - \exp(-\beta E_{\mathrm{lo}})}{U(T)}\phi(\lambda) \tag{5}$$

where the wavelength-integrated cross-section, σ is given by

$$\sigma = g_{\mathrm{lo}} f \frac{\pi e^2 \lambda_0^2}{m_e c^2} \quad . \tag{6}$$

Here, n is the number density of the line species, E_{up} and E_{lo} are the upper and lower energy levels of the transition, $\beta = 1/kT$ is the thermodynamic beta, ϕ is the normalized line profile, U is the species partition function, T is the temperature, f is the oscillator strength, g_{lo} is the degeneracy of the lower level, λ_0 is the line center, and m_e is the electron mass.

For all lines of all species besides hydrogen, ϕ is approximated with a Voigt profile using the numerical approximation from Hunger (1956). The width of the Gaussian component is

$$\sigma_D = \lambda_0 \sqrt{\frac{2kT}{m} + \xi^2} \quad , \tag{7}$$

where m is the species mass, and ξ is the microturbulent velocity, a fudge factor used to account for convective Doppler-shifts. The width of the Lorentz component (in frequency rather than wavelength units) is given by

$$\Gamma = \Gamma_{\mathrm{rad}} + \gamma_{\mathrm{stark}} n_e + \gamma_{\mathrm{vdW}} n_{\mathrm{H\ I}} \quad , \tag{8}$$

where Γ_{rad} is the radiative broadening parameter, γ_{stark} the per-electron Stark broadening parameter, γ_{vdW} the per-neutral-hydrogen van der Waals broadening parameter, and n_e and $n_{\mathrm{H\ I}}$ are the number densities of free electrons and neutral hydrogen, respectively. We neglect pressure broadening of molecular lines, setting γ_{stark} and γ_{vdW} to zero.

When Γ_{rad} is not supplied in the linelist, Korg approximates it with

$$\Gamma_{\mathrm{rad}} = \frac{8\pi^2 e^2}{mc\lambda^2} gf \quad , \tag{9}$$

where m is the mass of the atom or molecule, g is the degeneracy of the lower level of the transition, and f is the transition oscillator strength. This can be obtained by assuming that spontaneous deexitation dominates the transition's energy uncertainty, and that the upper level's degeneracy is unity.

The values of γ_{stark} and γ_{vdW} at 10^5 K, $\gamma_0^{\mathrm{stark}}$ and γ_0^{vdW}, are provided by the linelist, then scaled by their temperature dependence (per semiclassical impact theory) to the the per-particle broadening parameters:

$$\gamma_{\mathrm{stark}} = \gamma_0^{\mathrm{stark}} \left(\frac{T}{10^5\ \mathrm{K}}\right)^{\frac{1}{6}} \tag{10}$$

$$\gamma_{\mathrm{vdW}} = \gamma_0^{\mathrm{vdW}} \left(\frac{T}{10^5\ \mathrm{K}}\right)^{\frac{3}{10}} \tag{11}$$

Alternatively, Anstee, Barklem, and O'Mara (ABO; Anstee & O'Mara 1991) parameters, which describe a more nuanced temperature dependence in broadening by neutral hydrogen, in which case γ_{vdW} will be calculated according to that formalism. When a Stark broadening parameter is not provided in the linelist, the approximation from Cowley (1971) is used. When a van der Waals broadening parameter is provided, a form of the the Unsöld approximation (Unsold 1955; Warner 1967) is used in which the angular momentum quantum number is neglected and the mean square radius, $\overline{r^2}$, is approximated by

$$\overline{r^2} = \frac{5}{2}\left(\frac{n^{*2}}{Z}\right)^2 \quad , \tag{12}$$

where n^* is the effective principal quantum number and Z is the atomic number. Pressure broadening is neglected for autoionizing lines with no provided parameters.

Except for hydrogen lines, the absorption coefficient for each line is calculated over a dynamically determined wavelength range. The maximum detuning (wavelenth difference from the line center) is set to the value at which a pure Gaussian or pure Lorentzian profile takes on a value of α_{crit}, whichever is larger. By default, Korg truncates profiles at $\alpha_{\mathrm{crit}} = 10^{-3}\alpha_{\mathrm{cntm}}$, where α_{cntm} is the local continuum absorption coefficient. This ratio, $\alpha_{\mathrm{crit}}/\alpha_{\mathrm{cntm}}$, can be set by the user.

The broadening of hydrogen lines is treated separately. For H_α, H_β, and H_γ, we treat ϕ as a Voigt profile using the p-d approximation from Barklem et al. (2000). For other hydrogen lines, where Stark broadening dominates, we use the precomputed profiles from Stehlé & Hutcheon (1999), which include Stark and Doppler broadening.

2.4. *Continuum absorption*

Korg computes contributions to the continuum absorption coefficient from a number of sources, listed in Table 1. Continuum absorption mechanisms involving an atom and an electron can be classified as bound-free (bf) or free-free (ff), depending on whether the electron is initially bound to the atom. Note that these are named as though the electron were bound, even for free-free sources, so, e.g., H I bf refers to photons which ionize neutral hydrogen neutral hydrogen, and H I ff refers to photons absorbed by the interaction of free electrons and protons. The treatment of bf and ff absorption for hydrogenic atoms is taken from Kurucz (1970) (which is a transcription of Karzas & Latter (1961) for bf absorption) and van Hoof et al. (2014), and the remaining bf and ff sources are from polynomial fits provided by Gray (2005). Rayleigh scattering by H, He, and H_2 is also included (see Table 1 for citations). The contribution to the absorption coefficient from electron scattering, α_e, is given by

$$\alpha_e = \frac{8\pi}{3}\left(\frac{e^2}{m_e c^2}\right)^2 \tag{13}$$

(e.g. Gray 2005).

Korg also provides experimental support for metal continuum absorption using TOPbase data . . .

2.5. *Inputs*

Korg includes functions for parsing "Kurucz"-format model atmospheres, and line lists in the format supplied by VALD (Piskunov et al. 1995; Ryabchikova et al. 2015) or Kurucz (Kurucz 2011), or those accepted by MOOG. When parsing VALD linelists, it will automatically apply corrections to unscaled $\log(gf)$ values for transitions with isotope information using the isotopic abundance values supplied by Berglund & Wieser (2011). Korg

Absorption source	Energy bounds	Temperature bounds	Reference
H I bf			Kurucz (1970)
H I ff	100 Å – 10^6 Å	100 K – 10^6 K	van Hoof et al. (2014)
H^- bf	$\geq$ 1250 Å		Wishart (1979)*
H^- ff	2604 Å – 113,918 Å	2520 K – 10,080 K	Bell & Berrington (1987)*
H_2^+ bf and ff	700 Å – 20,000 Å	3150 K – 25,200 K	Stancil (1994)
He II bf			Kurucz (1970)
He II ff	100 Å – 10^6 Å	100 K – 10^6 K	van Hoof et al. (2014)
He^- ff	5063 Å – 151,878 Å	2520 K – 10,080 K	John (1994)*
H Rayleigh	$\geq$ 1300 Å		Lee (2005)†
He Rayleigh	$\geq$ 1300 Å		Dalgarno & Kingston (1960); Dalgarno (1962); Schwerdtfeger (2006)†
H_2 Rayleigh	$\geq$ 1300 Å		Dalgarno & Williams (1962)
electron scattering			see text

Table 1. Sources of continuum absorption in KORG. References marked with "*" are via polynomial fits in Gray (2005). Those marked with "†" are via Colgan et al. (2016).

detects and uses packed Anstee, Barklem, and O'Mara (ABO; Anstee & O'Mara 1991) if they are provided (as VALD optionally does). Users may also construct line lists or model atmospheres using custom code and pass them directly into KORG.

3. SCIENTIFIC VERIFICATION

To validate KORG, we compare its spectra to those produces by MOOG and TURBOSPECTRUM for four stars: the Sun, Arcturus, HD122563 (an extremely metal-poor red giant), and HD49933 (an F-type main sequence star). Figures 1 – 4 show these comparisons in six wavelength regions:

- 3660 Å – 3670 Å, near the Balmer jump
- 3935 Å – 3948 Å, in the wing of the Ca II Fraunhofer K line
- 5160 Å – 5175 Å, including two lines of the Fraunhofer b Mg triplet
- 6540 Å – 6580 Å, including H_α
- 15000 Å – 15050 Å, part of the APOGEE wavelength region
- 15000 Å – 15050 Å, in the near-infrared, part of the APOGEE wavelength region
- The continuum across 2000Å– 10000 Å, computed without any lines

For the first four regions, we used an "extract all" linelist from VALD, which includes all known lines in within th wavelength bounds. The near-infrared spectrum was synthesized using the APOGEE DR16 linelist. As MOOG can handle a limited number of lines within a 2 Å window, we culled the weakest TiO lines from each list before passing it to MOOG.

4. BENCHMARKS

Figure 5 shows the time taken by of KORG, TURBOSPECTRUM, and MOOG to produce the spectra in Section 3. The infrared region was extended to 15000 Å – 15500 Å to provide a example synthesis over a larger wavelength region. We note that this region, synthesized with the APOGEE DR16 linelist, provides a realistic example of synthesizing a spectrum from a large "industrial" survey. The others, which used VALD "extract all" linelists, include more lines than would typically used, including many which have essentially on effect on the spectrum. We note also that KORG only need to load linelists and model atmosphere into memory once, so repeat syntheses which use the same inputs (as would be the case when varying individual abundances) avoid that step. We note also that the MOOG values should not be taken at face value, since the linelists provided to MOOG were reduced in size by an order of magnitude or more.

In Figure 6 we plot the time required by KORG to compute gradient spectra with regard to a varying numbers of element abundances, as well as a subset of the derivative spectra. We used the near-infrared wavelength range and APOGEE linelist. A dashed horizontal line marks the time required to synthesize the spectrum, just under one second. Calculating the N-dimensional gradient spectrum takes roughly $2 + 0.15N$ seconds, meaning that the marginal time required for each derivative is an order of magnitude smaller than would be required to obtain it via finite differences. As JULIA's automatic differentiation ecosystem improves,

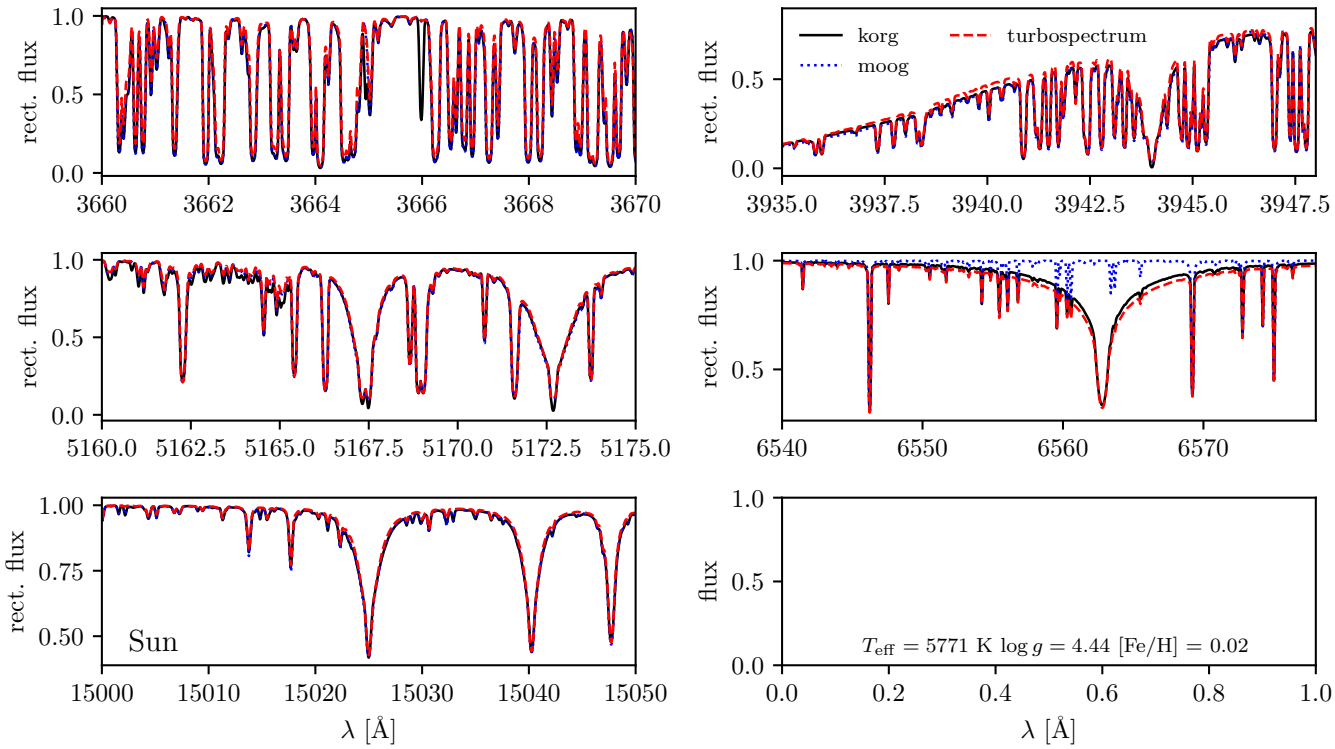

Figure 1. Portions of a synthetic solar spectrum generated with Korg, Moog, and Turbospectrum, as well as the continuum generated with an empty linelist. See text for details. Major discrepancies are mostly due to problems with model atmosphere interpolation and interpretation, which I am working on now.

Korg may benefit from further speedups to the calculation of gradient spectra.

5. CONCLUSIONS

We have presented a new code, Korg, for 1D LTE spectral synthesis. The code is publicly available at https://github.com/ajwheeler/Korg.jl and installable via the Julia package manager. Detailed documentation, usage examples (including the use of ForwardDiff cite to compute derivatives of the spectrum with regard to input parameters) , and installation instructions are available at https://ajwheeler.github.io/Korg.jl/stable.

ACKNOWLEDGMENTS

AJW would like to thank Chris Sneden for his advice, Samuel Potter, Paul Barklem, Karen Lind, and Thomas Nordlander for answers to naive questions, and Rob Rutten, David F. Gray, Ivan Hubeny, and Robert Kurucz for writing excellent reference material.

AJW is supported by the National Science Foundation Graduate Research Fellowship under Grant No. 1644869. MKN is in part supported by a Sloan Research Fellowship.

REFERENCES

Amarsi, A. M., Asplund, M., Collet, R., & Leenaarts, J. 2015, Mon. Not. R. Astron. Soc., 454, L11, doi: 10.1093/mnrasl/slv122

Amarsi, A. M., Lind, K., Osorio, Y., et al. 2020, Astron. Astrophys., 642, A62, doi: 10.1051/0004-6361/202038650

Anstee, S. D., & O'Mara, B. J. 1991, Monthly Notices of the Royal Astronomical Society, 253, 549, doi: 10.1093/mnras/253.3.549

Asplund, M., Grevesse, N., Sauval, A. J., & Scott, P. 2009, Annu. Rev. Astron. Amp Astrophys. Vol 47 Issue 1 Pp481-522, 47, 481, doi: 10.1146/annurev.astro.46.060407.145222

Barklem, P. S., & Collet, R. 2016, Astron. Astrophys., 588, A96, doi: 10.1051/0004-6361/201526961

Barklem, P. S., Piskunov, N., & O'Mara, B. J. 2000, ArXivastro-Ph0010022. https://arxiv.org/abs/astro-ph/0010022

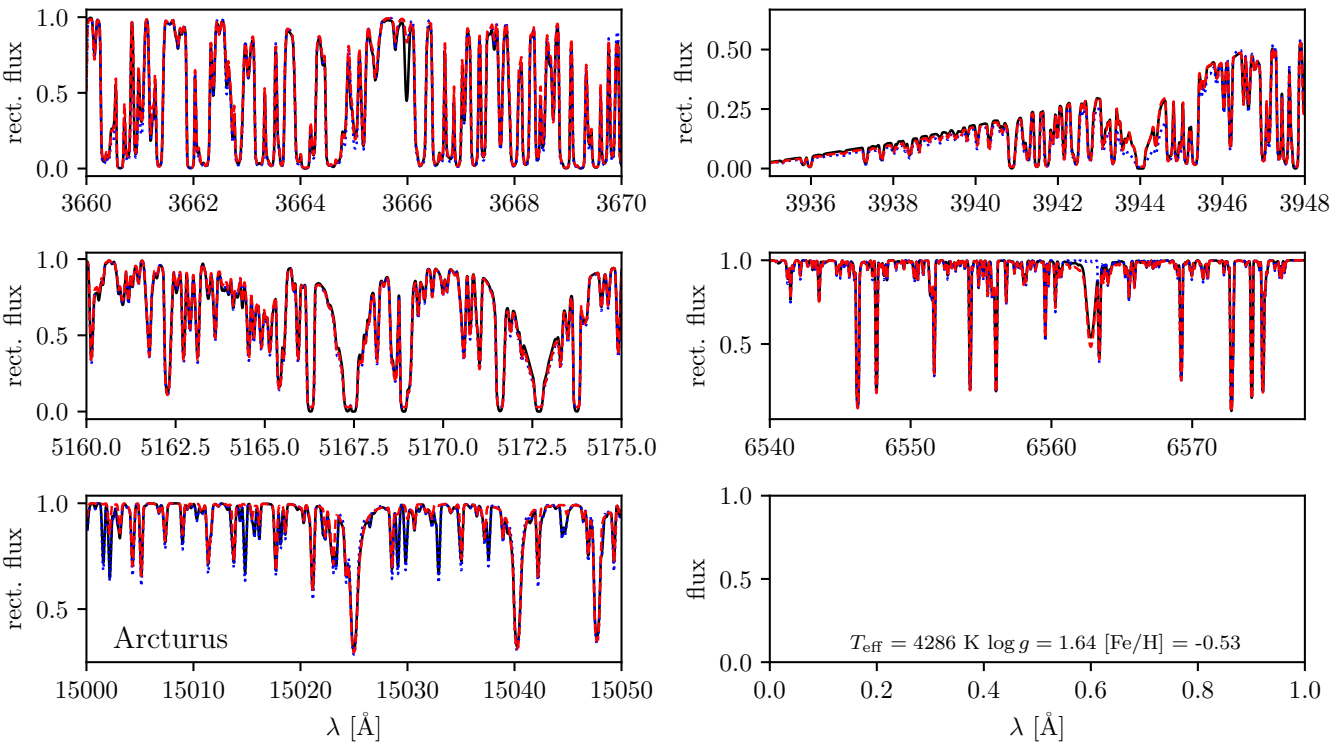

Figure 2. Same as Figure 1, but showing the synthetic spectrum of Arcturus. The "missing lines" in the infrared are due to lack of isotopic scaling. This is addressed in v6.0.

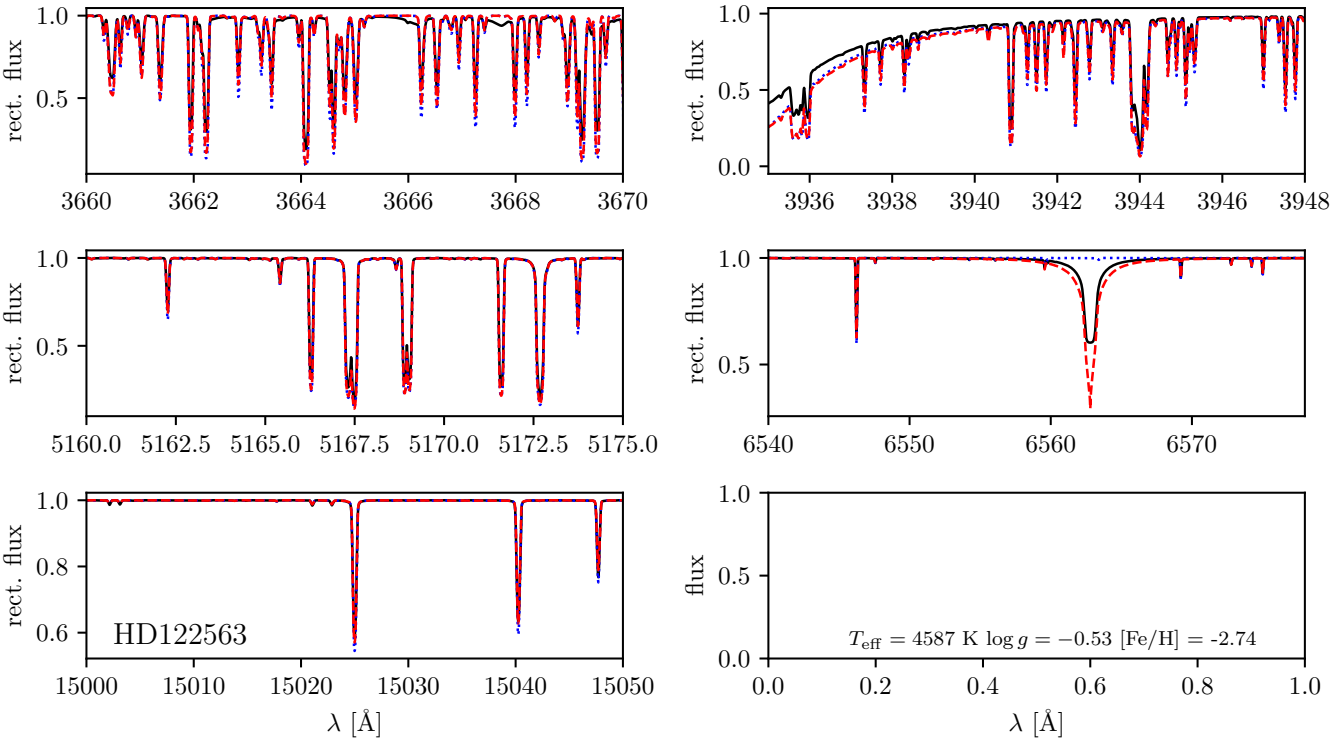

Figure 3. Same as Figure 1, but showing the synthetic spectrum of HD122563.

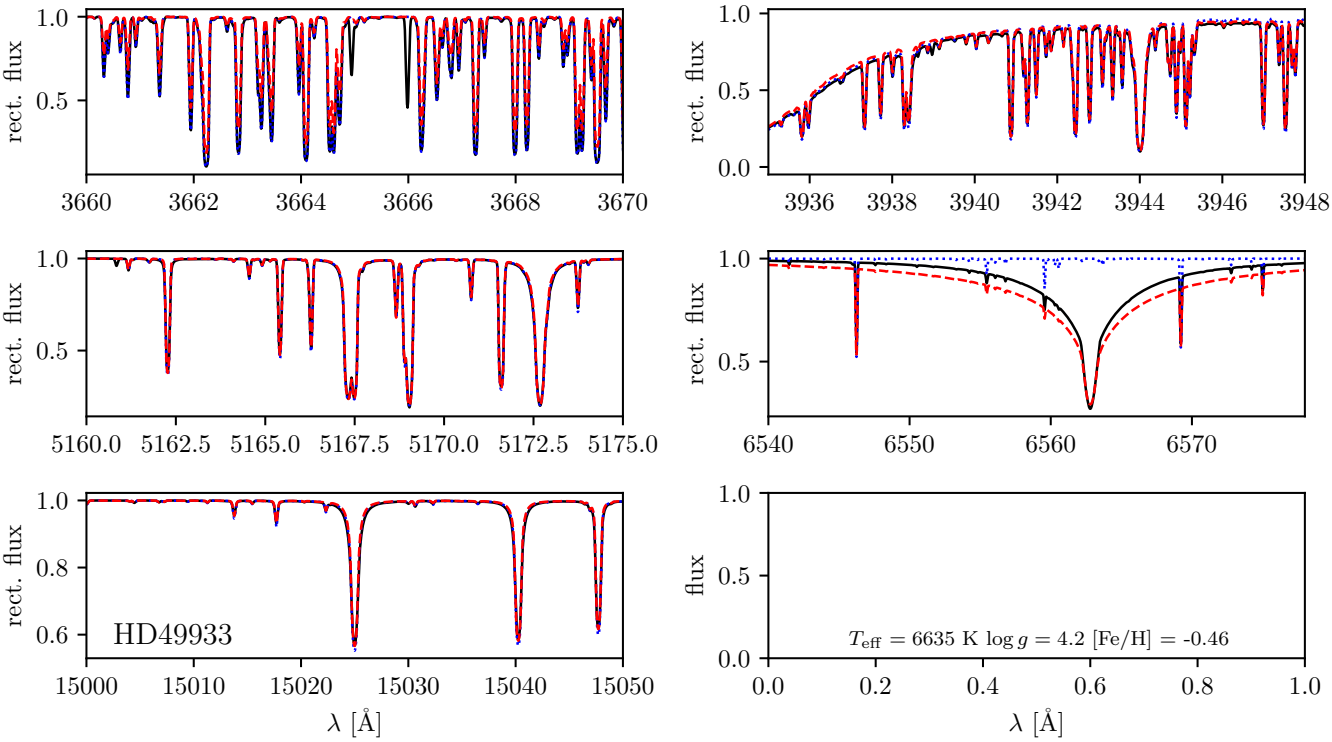

Figure 4. Same as Figure 1, but showing the synthetic spectrum of HD49933.

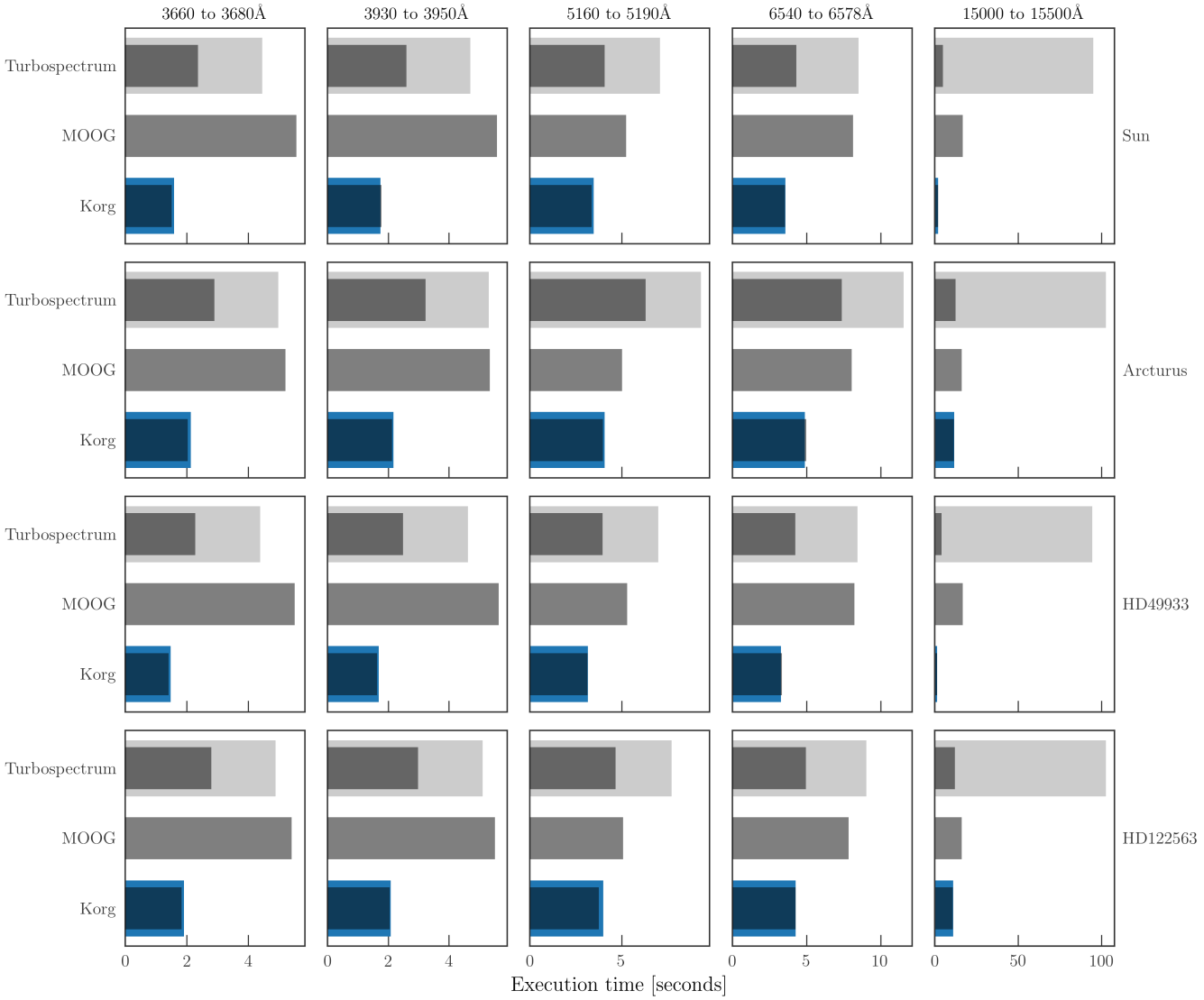

Figure 5. The time taken by KORG, MOOG, and TURBOSPECTRUM for each synthesis in Section 3. Note that the wavelength region synthesize were slightly larger than those plotted in Figures 1 – 4. The rightmost column, corresponding to the near-infrared spectrum synthesized from the the APOGEE DR17 linelist, most closely resembles real-world use.

Bell, K. L., & Berrington, K. A. 1987, J. Phys. B At. Mol. Phys., 20, 801, doi: 10.1088/0022-3700/20/4/019

Bergemann, M., Hansen, C. J., Bautista, M., & Ruchti, G. 2012, Astron. Astrophys., 546, A90, doi: 10.1051/0004-6361/201219406

Berglund, M., & Wieser, M. E. 2011, Pure Appl. Chem., 83, 397, doi: 10.1351/PAC-REP-10-06-02

Buder, S., Sharma, S., Kos, J., et al. 2021, Mon. Not. R. Astron. Soc., 506, 150, doi: 10.1093/mnras/stab1242

Colgan, J., Kilcrease, D. P., Magee, N. H., et al. 2016, Astrophys. J., 817, 116, doi: 10.3847/0004-637X/817/2/116

Cowley, C. R. 1971, The Observatory, 91, 139

Dalgarno, A. 1962, Spectral Reflectivity of the Earth's Atmosphere III: The Scattering of Light by Atomic Systems, Tech. Rep. 62-20-A, Geophysical Corporation of America

Dalgarno, A., & Kingston, A. E. 1960, Proc. R. Soc. Lond. Ser. A, 259, 424, doi: 10.1098/rspa.1960.0237

Dalgarno, A., & Williams, D. A. 1962, Astrophys. J., 136, 690, doi: 10.1086/147428

Gray, D. F. 2005, The Observation and Analysis of Stellar Photospheres

Gray, R. O., & Corbally, C. J. 1994, The Astronomical Journal, 107, 742, doi: 10.1086/116893

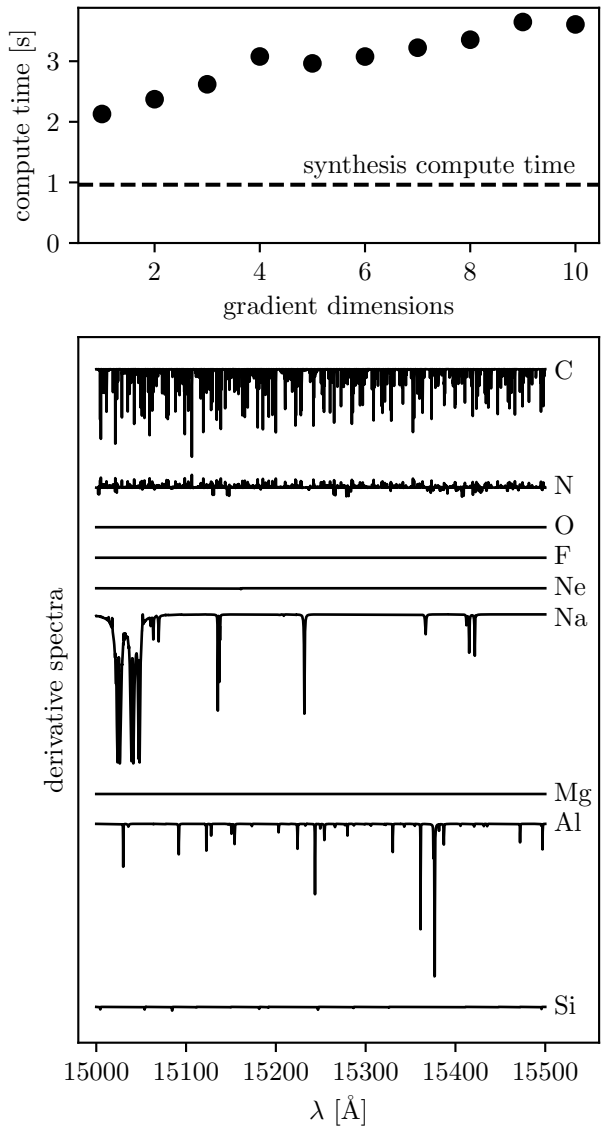

Figure 6. top: Compute time required to compute derivative of spectrum with regard to N element abundances, for varying N. **bottom:** The derivative spectra.

Haynes, W. M., ed. 2010, CRC Handbook of Chemistry and Physics, 91st Edition, 91st edn. (Boca Raton, Fla.: CRC Press)

Holtzman, J. A., Hasselquist, S., Shetrone, M., et al. 2018, AJ, 156, 125, doi: 10.3847/1538-3881/aad4f9

Hunger, K. 1956, Z. Astrophys., 39, 36

John, T. L. 1994, Mon. Not. R. Astron. Soc., 269, 871, doi: 10.1093/mnras/269.4.871

Karzas, W. J., & Latter, R. 1961, Astrophys. J. Suppl. Ser., 6, 167, doi: 10.1086/190063

Kurucz, R. L. 1970, SAO Spec. Rep., 309

—. 1993, Kurucz CD-ROM, Cambridge, MA: Smithsonian Astrophysical Observatory, —c1993, December 4, 1993

—. 2011, Can. J. Phys., 89, 417, doi: 10.1139/p10-104

Lee, H.-W. 2005, Mon. Not. R. Astron. Soc., 358, 1472, doi: 10.1111/j.1365-2966.2005.08859.x

Lind, K., Asplund, M., Barklem, P. S., & Belyaev, A. K. 2011, Astron. Astrophys., 528, A103, doi: 10.1051/0004-6361/201016095

Mogensen, P. K., Carlsson, K., Villemot, S., et al. 2020, JuliaNLSolvers/NLsolve.Jl: V4.5.1, Zenodo, doi: 10.5281/zenodo.4404703

Osorio, Y., & Barklem, P. S. 2016, Astron. Astrophys., 586, A120, doi: 10.1051/0004-6361/201526958

Piskunov, N. E., Kupka, F., Ryabchikova, T. A., Weiss, W. W., & Jeffery, C. S. 1995, Astron. Astrophys. Suppl. Ser., 112, 525

Plez, B. 2012, Astrophysics Source Code Library, ascl:1205.004

Plez, B., Smith, V. V., & Lambert, D. L. 1993, The Astrophysical Journal, 418, 812, doi: 10.1086/173438

Ryabchikova, T., Piskunov, N., Kurucz, R. L., et al. 2015, Phys. Scr., 90, 054005, doi: 10.1088/0031-8949/90/5/054005

Sbordone, L., Bonifacio, P., Castelli, F., & Kurucz, R. L. 2004, Memorie della Societa Astronomica Italiana Supplementi, 5, 93

Schwerdtfeger, P. 2006, in Computational, Numerical and Mathematical Methods in Sciences and Engineering, Vol. Volume 1, Atoms, Molecules and Clusters in Electric Fields (PUBLISHED BY IMPERIAL COLLEGE PRESS AND DISTRIBUTED BY WORLD SCIENTIFIC PUBLISHING CO.), 1–32, doi: 10.1142/9781860948862_0001

Sneden, C., Bean, J., Ivans, I., Lucatello, S., & Sobeck, J. 2012, Astrophysics Source Code Library, ascl:1202.009

Stancil, P. C. 1994, Astrophys. J., 430, 360, doi: 10.1086/174411

Stehlé, C., & Hutcheon, R. 1999, Astron. Astrophys. Suppl. Ser., 140, 93, doi: 10.1051/aas:1999118

Unsold, A. 1955, Physik Der Sternatmospharen, MIT Besonderer Berucksichtigung Der Sonne.

Valenti, J. A., & Piskunov, N. 1996, Astronomy and Astrophysics Supplement Series, 118, 595

—. 2012, Astrophysics Source Code Library, ascl:1202.013

van Hoof, P. A. M., Williams, R. J. R., Volk, K., et al. 2014, Mon. Not. R. Astron. Soc., 444, 420, doi: 10.1093/mnras/stu1438

Warner, B. 1967, . ., 136, 8

Wishart, A. W. 1979, Mon. Not. R. Astron. Soc., 187, 59P, doi: 10.1093/mnras/187.1.59P

APPENDIX

A. NUMERICAL EVALUATION OF TRANSFER INTEGRALS

To solve the equation of radiative transfer along a ray, KORG approximates the source function, S (λ subscripts dropped for brevity), with linear interpolation over optical depth, $\tau = \alpha \mathrm{d}s$. Between each adjacent pair of atmospheric layers, we have $S(\tau) \approx m\tau + b$, so the (indefinite) transfer integral has solution

$$\int (m\tau - b) \exp(-\tau) \mathrm{d}\tau = -\exp(-\tau)\left(b + m(\tau + 1)\right) \quad . \tag{A1}$$

When evaluating equation 4, KORG uses the same approximation,

$$\int (m\tau - b) E_2(\tau) \mathrm{d}\tau = \frac{1}{6}\left[\tau E_2(\tau)(3b + 2m\tau) - \exp(\tau)(3b + 2m(\tau + 1))\right] \quad . \tag{A2}$$